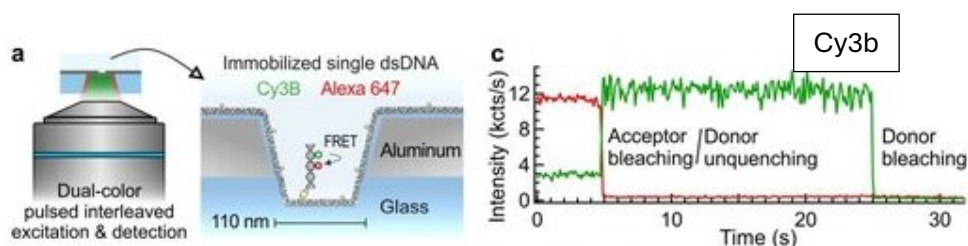


1. Draw E_{FRET} vs R in a XY-graph and mark Forster radius (R_0) with dotted lines. 3+1
2. The R_0 for Cy3b and Alexa647 dye-pair is 6.2 nm. Now see the following data for one event and estimate the corresponding donor-acceptor distance as labelled in the DNA. 6



3. You have setup a confocal/FCS microscope using an 60X water-objective with 1.2 NA. The refractive index of water is 1.33. For a laser of 488 nm, (a) find out the effective confocal volume in fL ($V_{eff} = \pi^{2/3} R_{xy}^2 R_z$). (b). If the concentration of the dye-labelled protein solution is 10 nM, find out the average number of dye-labelled protein molecules within the confocal volume at a time. 4 + 2 = 6
4. (a) A fluorophore switches between ON and OFF states with ON time = 10 ms and OFF time = 5 s. What is the average duty cycle in percentage? Duty cycle is the fraction of time a fluorophore is in the ON state. 2
(b) How many fluorophores per μm^2 can you image simultaneously if your system resolves ~ 1 fluorophore/ μm^2 at a time? 2
5. You have a $10 \times 10 \mu\text{m}$ field with a labelling density of 5000 fluorophores/ μm^2 . If your imaging setup can localize 50 fluorophores per frame and you run at 100 fps, estimate total acquisition time to reach full sampling. 2
6. Schematically depict the Jablonski Diagram using (i) Anharmonic Potential Energy diagrams for each electronic state and use arrows for transitions. (ii) Label X, Y axes. (iii) Mark the vibrational energy states at each electronic state. (iv) Show the time-scale of the transitions during absorption of photons and emission of photons. Show the time-scale for dissipative energy transitions, and (v) finally depict why fluorescence

emission is red-shifted than the absorbance. $2 + 1 + 2 + 3 + 2 = 10$

7. Short Answers-

- (i) What kind of fluorophores are used in PALM and STORM? **3**
- (ii) How does STED improve resolution beyond the diffraction limit? **2**
- (iii) Why is spectral overlap between donor emission and acceptor absorption essential in FRET. Explain pictorially, and how does it affect R_0 ? **2+1**
- (iv) Why do swimmers like bacteria need non-reciprocal (asymmetric) motion to move at low Reynolds number? **2**
- (v) What is the physical meaning of the viscous drag coefficient γ for a sphere in a fluid? How is it related to the sphere's size and the fluid's viscosity? **3**
- (vi) What is the diffusion constant of a 100 nm radius particle in water at room temperature (298 K). ($\eta=10^{-3}$ Pa.s and $k_B = 1.38 \times 10^{-23}$ J/K). **3**
- (vii) If the viscosity of the fluid is doubled to 0.002 Pa.s while keeping the particle size and shape constant, calculate the new viscous friction coefficient for a particle with a radius of 2 μm . **2**